

DARPAN NEMADE

Ahmednagar, Maharashtra | +91 9356606402 | darpannemade2004@gmail.com
LinkedIn | GitHub

Summary

AI/ML Engineer skilled in building machine learning pipelines, LLM-powered applications, RAG systems, and scalable automation workflows using n8n. Experienced in model training, feature engineering, evaluation, and integrating AI models into real-world systems. Strong problem-solving, analytical thinking, and collaboration skills with a track record of delivering high-impact ML and GenAI solutions.

Technical Skills

Machine Learning: Supervised/Unsupervised Learning, Feature Engineering, Model Evaluation, Validation

GenAI & NLP: LLaMA 3, RAG, Prompt Engineering, Agentic Workflows, Embeddings, Summarization

Deep Learning: PyTorch, TensorFlow, CNNs, RNNs, Transformers, Training Pipelines

Automation & Pipelines: n8n Automation, Workflow Orchestration, REST Webhooks, ETL-style Pipelines

Computer Vision: YOLOv8, Faster R-CNN, OpenCV, UE5 Integration

Development Tools: Streamlit, Flask, MongoDB, WebSockets, Solidity, CUDA

Programming: Python, JavaScript, Java

Projects

ODIN — AI-Powered NFT Marketplace (link) Aug 2025

- Built an LLM-based wallet agent for secure blockchain actions (minting, transfers, balance queries), reducing user interaction effort by 55%.
- Implemented an ML-driven RAG retrieval layer using embeddings for accurate intent classification and context-aware recommendations.
- Designed n8n-based automation workflows for transaction monitoring, event triggers, and Slack/Trello notifications.
- Added EIP-191 signature verification lowering unauthorized wallet activity by 40%.
- Performed gas usage benchmarking and optimized Solidity contracts achieving an 18% improvement in efficiency.

EvoCine — Multimodal Cinematic Trailer Generator Sep 2025

- Engineered a text-to-video ML pipeline combining SDXL, AnimateDiff, and Stable Video Diffusion for multimodal scene generation.
- Applied CLIP-based scoring with genetic optimization improving aesthetic and narrative coherence by 40%.
- Integrated Coqui TTS, MusicGen, and Wav2Lip to align audio, lip movement, and generated scenes.
- Applied Real-ESRGAN upscaling and frame refinement to produce stable 1080p/2K-quality output.

Thermal Surveillance System (YOLOv8 + Faster R-CNN)(link) Apr 2025

- Built a two-stage computer vision pipeline using YOLOv8 and Faster R-CNN improving detection accuracy on thermal footage by 32%.
- Performed preprocessing, augmentation, and evaluation using mAP and IoU metrics.
- Achieved real-time inference (<120ms) using CUDA optimization and asynchronous processing.
- Integrated UE5 visualization for rendering 3D bounding overlays from live detections.

Music Streaming Platform(link) Apr 2025

- Developed JWT-secured REST APIs reducing unauthorized access attempts by 25%.
- Delivered low-latency audio streaming (<50ms) using WebSockets and optimized buffering.
- Improved MongoDB query efficiency by 35% using aggregation pipelines and indexing.

Certifications

IBM Generative AI Certification — IBMCE

MongoDB Full-Stack Developer Certification — MongoDB University

Education

Bachelor of Technology, VIT Bhopal University

Expected 2026

Specialization: Computer Science (AI & ML) — GPA: 8.61/10

Leadership

House Captain — Yashshree Academy

2017–2020

Event Committee Member — VIT Maharashtra Club

2023–2025